Reading Notes and Reproduction of Closed-Form Concept Erasure via Double Projections

Object Erasure: Cassette Player on Stable Diffusion 1.4

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Overview

This report presents a technical reading of Zhang et al. [1] on Double Projections (DP) for concept erasure, together with a reproduction of the Stable Diffusion 1.4 object-erasure run for *cassette player* (guide: *box*) on a lab RTX 5090. In the reproduction, ResNet-50 semantic accuracy on the target fell from 76.0% to 2.0% (authors' reference: 78% → 2%), and the average drop on the other nine ImageNette classes was 2.8% (reference: 3.3%). Section 1 develops the method with its mathematical detail (UCE baseline, DP projections, truncation theory); Section 2 situates DP among closed-form editing methods; Sections 3–4 report the reproduction; Section 5 discusses open problems and directions.

1 Method (technical)

1.1 Problem and UCE baseline

Concept erasure edits a pretrained map $W_0 \in \mathbb{R}^{p\times n}$ (in practice, the UNet cross-attention `to_k` / `to_v` maps) so a target concept embedding c_i is redirected toward a proxy c_i^*, while non-target embeddings in $C_{\text{pres}} = [c_1, \dots, c_m]$ should stay close to their original images under W_0.

Unified Concept Editing (UCE) [2] casts this as a single least-squares objective over an edit set E and a preserve set P:

$$\min_W \sum_{c_i\in E} \left\|Wc_i - v_i^*\right\|_2^2 + \sum_{c_j\in P} \left\|Wc_j - W_0c_j\right\|_2^2, \qquad v_i^* := W_0\, c_i^*, \tag{1}$$

i.e., each target is pushed toward the image of its proxy under the *original* map, while preserve concepts are kept close to their original outputs. Setting the gradient to zero yields the closed form

$$W \;=\; \Big(\sum_{c_i\in E} v_i^* c_i^\top + \sum_{c_j\in P} W_0 c_j c_j^\top\Big)\Big(\sum_{c_i\in E} c_i c_i^\top + \sum_{c_j\in P} c_j c_j^\top\Big)^{-1}, \tag{2}$$

a deterministic, training-free update computed purely from concept embeddings (paper Eqs. (5)–(6)).

Why the least-squares fit still leaks (paper Theorem 3.1). A closed form does not by itself control collateral damage. For a single target c edited to v^*, with $N := cc^\top + C_{\text{pres}}C_{\text{pres}}^\top$,

the UCE update is the rank-one step

$$\Delta W \;=\; \big(v^* - W_0 c\big)\,\frac{c^\top N^{-1}}{c^\top N^{-1} c}, \tag{3}$$

which modifies the weights exactly along the residual direction $v^* - W_0 c$. The perturbation of a preserve vector p is therefore parallel to that of the target itself, scaled by the $N^{-1/2}$-weighted inner product ratio $|c^\top N^{-1} p|/|c^\top N^{-1} c|$. Under the paper's assumption that, for some $\lambda > 0$, $\langle N^{-1/2}c, N^{-1/2}p\rangle \geq \lambda\,\langle N^{-1/2}c, N^{-1/2}c\rangle$, one obtains

$$\big\|\Delta W\, p\big\|_2 \;\geq\; \lambda\,\big\|\Delta W\, c\big\|_2. \tag{4}$$

In words: if a preserve vector is sufficiently aligned with the target in the $N^{-1/2}$-weighted geometry, the edit perturbs it at least as much as it perturbs the target. Optimality of a least-squares fit therefore says nothing about *which directions absorb the change*; DP's design instead makes preservation a hard constraint rather than a soft penalty.

1.2 DP formulation

DP writes $W = W_0 + \Delta W$ and separates erasure from preservation with two closed-form projections.

Projection 1 (proxy in a safe subspace). Let $S \in \mathbb{R}^{n\times k}$ be a basis for a safe subspace (columns are guide / safe concept directions). The proxy is the orthogonal projection of the target embedding onto that subspace:

$$c_i^* \;=\; S\,(S^\top S)^+\,S^\top c_i, \tag{5}$$

where $(\cdot)^+$ is the Moore–Penrose pseudoinverse. For the Cassette Player object setup in the official code / reference log, the guide is *box*. When $k = 1$, this step reduces to the usual single-guide UCE-style proxy.

Projection 2 (nullspace constraint; essential). Require that the weight update does not move preserve embeddings:

$$\Delta W\, C_{\text{pres}} \;=\; 0. \tag{6}$$

With that constraint, erasure is a linearly constrained least-squares problem

$$\min_{\Delta W} \big\|(W_0 + \Delta W)\, c_i - W_0\, c_i^*\big\|_2^2 \quad \text{s.t.} \quad \Delta W\, C_{\text{pres}} = 0. \tag{7}$$

Let $U_2 \in \mathbb{R}^{n\times(n-r)}$ be an orthonormal basis for the left nullspace of C_{pres} ($r = \operatorname{rank} C_{\text{pres}}$, so $U_2^\top C_{\text{pres}} = 0$). Feasible updates are exactly $\Delta W = Z U_2^\top$ for some $Z \in \mathbb{R}^{p\times(n-r)}$. Substituting into (7) removes the constraint and reduces the problem to

$$\min_{Z} \big\|Z\,x - b\big\|_2^2, \qquad x := U_2^\top c_i \in \mathbb{R}^{n-r}, \qquad b := W_0\,(c_i^* - c_i), \tag{8}$$

whose minimum-norm solution (for $x \neq 0$) is $Z = b\,x^\top/\|x\|_2^2$, giving the closed form

$$\Delta W \;=\; W_0\,(c_i^* - c_i)\,\frac{(U_2^\top c_i)^\top}{\|U_2^\top c_i\|_2^2}\,U_2^\top \tag{9}$$

(paper Eqs. (11)–(12)). The paper notes that multiple-target erasure $C_{\text{tgt}} = [c_1, \ldots, c_T]$ is solved identically with a matrix $X = U_2^\top C_{\text{tgt}}$.

Guarantee on C_{pres}. If $\Delta W C_{\text{pres}} = 0$ and $W^\star = W_0 + \Delta W$, then

$$W^\star v \;=\; W_0 v \quad \text{for all } v \in \text{col}(C_{\text{pres}}). \tag{10}$$

That is exact preservation of the linear map on the preserve embedding subspace (paper Theorem 4.1). It does *not* by itself guarantee identical ResNet accuracies on sampled images, because generation and classification are nonlinear and noisy.

Both projections are analytical: once embeddings are collected, the edit is deterministic and training-free. Projection 1 is optional (one can hand-pick c_i^* as in UCE); Projection 2 is the geometric constraint that makes preservation explicit.

1.3 Truncated preserve sets: preservation and erasure bounds

For large preserve sets, one keeps only the top-k left singular directions of C_{pres}. Write the thin SVD $C_{\text{pres}} = U_1 \Sigma V^\top$ with $U_1 = [u_1, \ldots, u_r]$ ordered by $\sigma_1 \geq \cdots \geq \sigma_r > 0$, let $U_{1,k} = [u_1, \ldots, u_k]$, and let $U_{2,k} \in \mathbb{R}^{n \times (n-k)}$ be an orthonormal basis of $\text{span}(U_{1,k})^\perp$. Parameterizing $\Delta W = Z U_{2,k}^\top$ enforces $\Delta W \, U_{1,k} = 0$: the top-k preserve directions are still preserved exactly, and (9) applies with $U_2 \mapsto U_{2,k}$.

Theorem 4.2 (preservation bound). For any preserve column p_i of C_{pres},

$$\left\| W^\star p_i - W_0 p_i \right\|_2 \;\leq\; \sigma_{k+1}(C_{\text{pres}}) \left\| Z^\star \right\|_2, \tag{11}$$

where σ_{k+1} is the $(k{+}1)$-st singular value of C_{pres}: the perturbation of a preserved column is controlled by the *tail* of the spectrum cut off by truncation, scaled by the size of the update. At $k = r$ the right-hand side vanishes and Theorem 4.1 is recovered.

Theorem 4.3 (erasure lower bound). The paper complements this with a matching lower bound on the target side: every target whose embedding has non-negligible projection onto the edited row space $\text{row}(U_{2,k}^\top C_{\text{tgt}})$ is guaranteed to be modified, and in the single-target case with $c \notin \text{col}(U_{1,k})$ the pseudoinverse solution achieves the exact fit $Wc = W_0 c^\star$.

1.4 Embedding-level preservation vs. generation-level accuracy

Theorems 4.1–4.3 are stated on content embeddings. In the SD pipeline the UNet cross-attention actually consumes fused tokens

$$z_j \;=\; c_j + q_j, \tag{12}$$

where q_j is a positional embedding, and the CLIP text encoder self-attention mixes tokens before they reach the UNet. Consequently $\Delta W z_j = \Delta W q_j \neq 0$ even when $\Delta W c_j = 0$: the edit still moves preserve-token keys/values through their positional component, and the cross-attention softmax turns that into sample-level deviations. Perfect linear preservation is therefore compatible with nonzero generation-level accuracy drops—consistent with the *gas pump* drop observed in this reproduction (74% → 58%, Table 3) and with the paper's Appendix C.

1.5 How object erasure is measured

On ImageNette-style object erasure with SD 1.4, images are generated from the original and erased models, then scored with a pretrained ResNet-50 (Top-1). *Semantic* Acc treats near-duplicate labels (e.g. cassette / tape player / radio) as a match; *strict* Acc requires the exact label. Lower target Acc after the edit means stronger erasure; lower average drop on the other classes means better preservation under this proxy metric.

2 Related work

DP belongs to a line of post-hoc, training-free edits of generative model weights. TIME [3] introduced a closed-form projection edit for the cross-attention key/value matrices, regularized by a global Frobenius term toward the original weights. UCE [2] generalized this to a least-squares objective that edits a set of concepts while softly preserving a designated concept set, and MEMIT [6] showed how the same family of closed-form updates can be constrained so that previously stored associations remain exactly satisfied. On the fine-tuning side, ESD [5] erases concepts by training the denoiser toward a negatively guided teacher, and concept ablation [4] fine-tunes the model to match an anchor concept on target prompts. DP combines the closed-form efficiency of these methods with an explicit nullspace preservation constraint and, unlike the fine-tuning approaches, comes with analytical preservation guarantees (Theorems 4.1–4.3).

3 Reproduction setup

Table 1: Hardware and software for the Cassette Player run.

Item	Value
Host	`gpu-user0`
Conda env	`DP` (PyTorch 2.10.0+cu128)
GPU	Logical GPU 1, RTX 5090 (`CUDA_VISIBLE_DEVICES=1`)
Base model	Local lean fp16 SD 1.4 (`variant="fp16"`)
Target / guide	cassette player / box
Preserve set C_{pres}	9 other ImageNette classes

Pipeline (official DP code):

1. Generate original SD 1.4 images (1000 samples, $\sim$24 m 39 s).
2. Run the closed-form DP edit (seconds).
3. Generate erased images (1000 samples, $\sim$25 m 18 s).
4. Score both sets with ResNet-50 semantic Top-1.

Deviations. The server could not reach Hugging Face, so a local lean fp16 checkpoint was used with `variant="fp16"` at loading time. Without that flag, loading failed (missing full-precision VAE weights). Safety-checker weights were not included in the lean package, so the checker was disabled for this research run. The linear solve still runs in float32 while the loaded weights are fp16, matching the provided code path.

4 Results

Table 2 compares the reproduced ResNet-50 numbers to the authors' Cassette Player DP reference log. Erased semantic Acc matches at 2.0%. Other-class average drop is 2.8% vs. 3.3% in the reference. Original Acc differs slightly (76% vs. 78%); likely contributors include the lean fp16 checkpoint, environment / diffusers version, and sampling variation (these were not ablated separately).

Table 2: ResNet-50 accuracy: this reproduction vs. the authors' Cassette Player DP reference.

Metric	Ours	Reference
Target Orig Acc (semantic)	76.0%	78.0%
Target Erased Acc (semantic)	**2.0%**	**2.0%**
Target Drop (semantic)	74.0%	76.0%
Target Orig Acc (strict)	4.0%	6.0%
Target Erased Acc (strict)	**1.0%**	**1.0%**
Other Orig Acc (avg)	86.3%	86.8%
Other Erased Acc (avg)	83.6%	83.4%
Other Average Drop	**2.8%**	3.3%

Per-class semantic Acc from the same eval (Table 3 excerpt): the target collapses, but preservation is uneven—e.g. *gas pump* drops from 74% to 58% (16 pp) even though it is inside C_{pres}. Average other-class drop therefore hides class-level variation.

Table 3: Selected per-class semantic Acc (%) from this reproduction (100 images/class).

Class	Original	Erased	Drop (pp)
cassette player	76	2	74
gas pump	74	58	16
english springer	94	86	8
church	86	83	3
tench	73	83	−10

Figure 1 shows original vs. erased samples from the same 2×1000 generated set. Figure 2 shows Acc and Acc-drop across all ten classes. Figure 3 restricts to *cassette player* prompts: ResNet Top-1 labels move away from cassette/tape/radio toward other categories after the edit (guide = *box* in this setup).



Figure 1: Left: original SD 1.4. Right: after DP erase of *cassette player* (guide = *box*). Rows 1–2: target prompts. Row 3: non-target *church*.

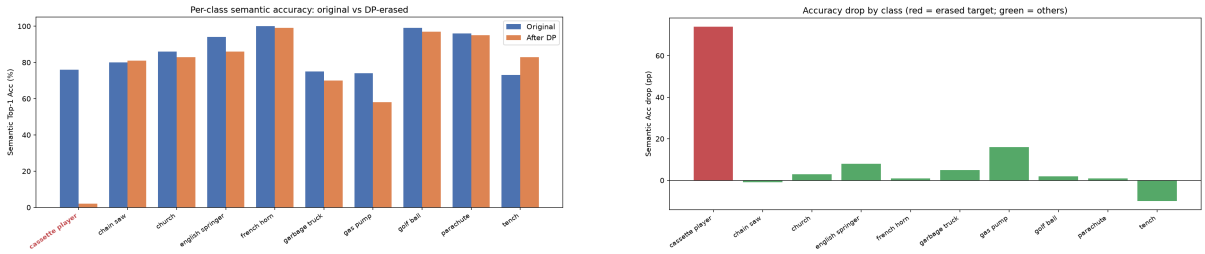


Figure 2: Left: semantic Top-1 Acc (%) before vs. after DP. Right: Acc drop (pp); red = target, green = preserve classes.

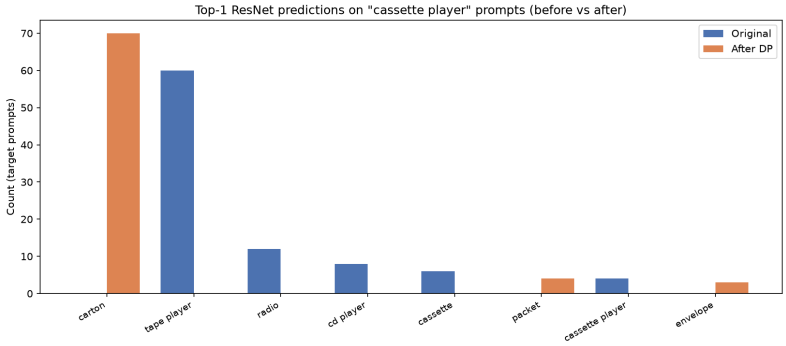


Figure 3: ResNet Top-1 labels on *cassette player* prompts only, before vs. after erasure.

5 Research gaps

Each gap below is stated together with a conjecture and a falsifiable test. Truncation, FLUX, and positional ablations were not re-run in this work; the proposed tests all run on the existing pipeline.

5.1 From the paper

1. **Truncated C_{pres}: how to pick k (Sec. 4.4, Thms. 4.2–4.3).** Theorem 4.2 bounds preserve-side damage by $\sigma_{k+1}\|Z^\star\|_2$, so the tradeoff is readable from the spectrum of C_{pres}. *Conjecture:* choosing k at a spectral gap of C_{pres} (a knee in $\sigma_1 \geq \cdots \geq \sigma_r$) should minimize worst-case preserve damage at fixed erasure strength. *Test:* sweep k on cassette player and check whether the knee in the σ-curve predicts the best operating point (max per-class drop vs. target Acc).

2. **Positional coupling (Appendix C).** Eq. (12) shows the guarantee holds on c_j, while the model consumes $z_j = c_j + q_j$. *Conjecture:* augmenting C_{pres} with position-augmented embeddings z_j (or editing the text-encoder embedding layer, as the paper does on FLUX) should shrink gas-pump-style drops. *Test:* re-run cassette player with C_{pres} built from z_j and compare per-class drops.

3. **Generalization beyond C_{pres} (Appendix J).** The guarantee covers only in-matrix concepts. *Conjecture:* for a held-out concept $p \notin C_{\text{pres}}$, the Theorem 3.1 mechanism reapplies—its drop should scale with its alignment to the target in the $N^{-1/2}$-weighted geometry. *Test:* measure Acc drop on a held-out class not listed in C_{pres} and correlate with $\langle N^{-1/2}c, N^{-1/2}p\rangle$.

5.2 From this reproduction

1. **Guide / proxy choice is under-specified.** Projection 1 needs a safe subspace S (here, guide *box*). Target Acc falling to 2% does not say what the model generates instead; Figure 3 shows Top-1 mass on target prompts moving to other labels after the edit. *Conjecture:* an acceptable substitute is one whose erased-image embeddings stay in a validated neighbourhood of the guide. *Proposal:* a *substitute audit*—report the Top-1 label distribution on target prompts plus CLIP text–image similarity to the guide—before declaring an erasure safe. For clinical use the guide would be generic anatomy and the audit would be clinician-reviewed, since a classifier histogram cannot catch a medically misleading substitute.

2. **Embedding-level preservation $\neq$ generation-level Acc.** Eq. (10) (paper Theorem 4.1) protects W on $\mathrm{col}(C_{\mathrm{pres}})$; in this reproduction, mean other-class drop is only 2.8%, but *gas pump* drops 16 pp (Table 3) inside C_{pres}. This is consistent with the positional-coupling mechanism of (12), not a contradiction of Theorem 4.1. *Interpretation:* theory and metric measure different objects, so both should be reported, and generation-level drops should be read as the noise floor of the linear guarantee.

3. **Mean drop hides per-class variation.** Average drop across nine preserve classes can look good while individual classes move a lot (Table 3). *Proposal:* report worst-class drop and the full per-class table as primary, mean drop as secondary.

4. **Clinical / safety transfer.** The geometric idea—edit only along directions to remove, constrained away from directions to keep—maps naturally onto medical generative safety: erase private or unsafe attributes while preserving diagnostic content. The missing pieces are domain guides (validated anatomy concepts), preserve sets from medical ontologies, and audit metrics beyond classification accuracy; identifying these is a direction for further study.

References

References

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[3] H. Orgad, B. Kawar, and Y. Belinkov. *Editing Implicit Assumptions in Text-to-Image Diffusion Models* (TIME). NeurIPS 2023.

[4] N. Kumari, B. Zhang, S.-Y. Wang, E. Shechtman, R. Zhang, and J.-Y. Zhu. *Ablating Concepts in Text-to-Image Diffusion Models*. ICCV 2023.

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Reproduction details: SD 1.4, *cassette player* (guide *box*), lab GPU (`gpu-user0`, RTX 5090, logical GPU 1), official DP code building on Unified Concept Editing.